SYSTEM ARCHITECTURE & STEP-BY-STEP METHODOLOGY

To build a high-fidelity, large-scale dataset of at least 20,000 sequences without propagating index-drift errors or data-imbalance anomalies, we introduce a **Cascading Multi-Agent Inline Boundary Tagging and Audit Engine**.
This architecture translates the annotation task from an unguided coordinate-generation problem into a deterministic text-rewriting and verification cycle. By decoupling the language models from direct mathematical index calculations and utilizing a cross-family double-agent layout, we optimize data integrity while strictly minimizing human annotation overhead.

1. Comprehensive Breakdown of Pipeline Stages

Step 1: Input Ingestion and Dataset Streaming

The pipeline streams unstructured consumer data from the source database one sequence at a time ($t \in D$). Operating at an individual row level ensures micro-granular resource management, preventing memory exhaustion when scaling up to the 20,000-row target boundary.
Before payload delivery, leading and trailing whitespace anomalies are programmatically corrected using standardized regular expression wrappers to establish a pristine string baseline for down-stream spatial alignment.

Step 2: Context-Aware In-Line Tagging (gemini-3.5-flash)

The raw string sequence is forwarded directly to gemini-3.5-flash, which operates as our **Primary Text-Rewriting Engine**. Unlike traditional sequence taggers that extract sub-token slices or output arbitrary coordinate vectors, this model is instructed to return the entire text verbatim while natively injecting custom XML/Markdown delimiters directly around target entities.

- **Linguistic Enforcement:** The model is governed by our predefined core categories: <MINOR_AGE>, <MINOR_EDU>, <GEN_NOUN>, <GEN_PHYS>, and <FAM_KIN>.
- **Semantic Boundary Filtering:** Operating at its default medium reasoning capacity, the model enforces the *Human Child Constraint* (filtering out pet ages and fictional personas common in e-commerce noise) and the *Reviewer Anchor Rule* (filtering out hypothetical buyer profiles) (Asiri et al., 2025; Mancera et al., 2025).
- **The Character Preservation Mandate:** The model is strictly constrained from altering capitalization, punctuation, or spelling. This guarantees that the original semantic structure remains completely intact during the rewrite ($t \rightarrow t_{tagged}$).

Step 3: Independent Cross-Family Guard Auditing (gpt-5.4-mini)

Once t_{tagged} is generated, it enters the **Automated Verification Node**. To eliminate systemic algorithmic bias, the pipeline switches ecosystems to OpenAI's gpt-5.4-mini. Passing the data to a model with a completely separate pre-training lineage, token-weight architecture, and context-optimization routine ensures that the auditor does not inherit the same contextual blind spots as the annotator. This split directly breaks potential collusive confirmation bias that naturally occurs within single-model validation environments (Liang et al., 2023).

The auditor is fed both the pristine ground-truth RAW_TEXT (t) and the rewritten ANNOTATED_TEXT (t_{tagged}). It executes a multi-point structural review using native JSON schema validation constraints:

- **Text Mutation Check:** It runs an exact sequence verification. If the annotator added, deleted, or accidentally modified even a single original character (e.g., expanding "12yo" into "12-year-old"), the row is immediately flagged.
- **Semantic Constraint Validation:** It audits whether the applied tags violate core boundaries (e.g., catching if the first model mistakenly tagged a pet's age or an abstract persona).
- **Structured Output Packaging:** If an error is detected, gpt-5.4-mini issues a structural FAIL verdict, automatically logs an explicit error_type category, and compiles a concise, one-sentence auditor_reason explaining the failure.

Step 4: The Decision Fork & Deterministic Regex Index Parsing

The orchestrator reads the validated JSON output from the auditor and splits the data flow:

Path A: The PASS Route (Automated Index Generation)

If the auditor returns a `PASS` status, the tagged text is immediately routed to a non-AI, deterministic regular expression parser. The parser uses compiled pattern matching (`re.finditer`) to sweep the text sequentially from left to right. Every time it hits an open tag, it grabs the label type, captures the exact literal text within the boundaries, slices out the XML tags, and records the absolute `start` and `end` character index coordinates relative to the progressively built clean text string. Because this counting is mathematical and sequential, it completely bypasses the index-indirection anomalies and off-by-one boundary shifts typical of generative models that reason over abstract sub-token weights instead of absolute character offsets (Mielke et al., 2021; Shin & Kaneko, 2024). The final payload is automatically committed to your production Postgres database as gold-standard training data.

Path B: The FAIL Route (Human-in-the-Loop Adjudication Queue)

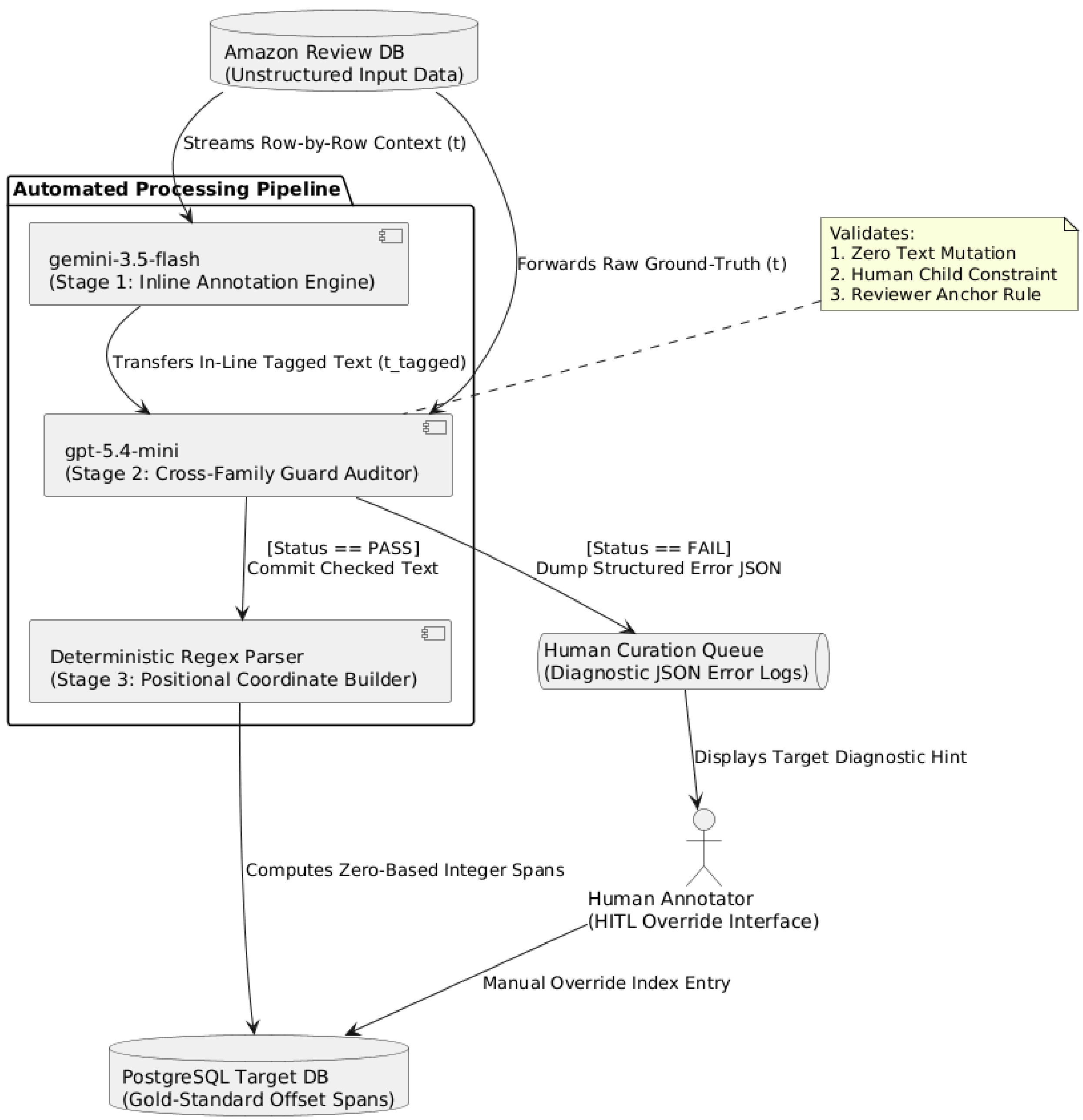
If the auditor returns a FAIL status, the pipeline halts automation for that record. The system logs the raw text, the faulty annotation, the error category, and the auditor's natural-language reason directly into an isolated human review queue.

Step 5: Targeted Human-in-the-Loop (HITL) Adjudication

Human domain experts access the failed rows through a dedicated user interface. Because the interface prominently displays the auditor's specific diagnostic hint (e.g., *"Model modified character structure '12yo' to '12 year old'"*), human annotators do not waste time re-reading the entire review to guess where the models disagreed. The reviewer simply checks the isolated error point, performs a manual override, and commits the finalized, error-free indices directly to the gold-standard database. This targeted mitigation workflow reduces human interaction to a tiny fraction of the total dataset, scaling processing speeds exponentially across your 20,000+ rows. Distributed agent workflows structured around explicit multi-stage validation tasks drastically minimize manual curation costs over broad data pools (Hegazy et al., 2025).

2. Formalized Manuscript Layout (Methodology Section)

Figure X: Multi-Agent Human-in-the-Loop Privacy NER Pipeline



A. Core Mathematical Framework

Let an input corpus sequence be represented as a vector of characters t. The transformation pipeline utilizes a generative rewriting operator, $G(Gemini)$, to produce a modified sequence containing inline semantic tags, t_{tagged}, such that $G(Gemini) \rightarrow t_{tagged}$.

To preserve the underlying mathematical utility of the textual sequences for downstream deployment, we enforce a strict invariant constraint:

$S(t) \equiv S(StripTags(t_{tagged}))$

where S represents the sequential character identity of the string block. Any mutation where the inner sequence character length diverges $\Delta S \neq 0$ is programmatically rejected by the secondary adversarial auditor, $A(GPT)$, forcing an immediate routing to human mitigation vectors.

B. Exhaustive Model Justification and Comparison Matrix

The model layout was chosen following a rigorous assessment of instruction-following fidelity and cost efficiency. While expensive frontier models provide high contextual reasoning, processing a dataset of 20,000+ rows twice (extraction + audit) introduces substantial API costs. Our tiered approach balances advanced model intelligence with lean operational costs:

- gemini-3.5-flash **(The Primary Extractor):** Chosen for Stage 1 because of its exceptional context-retention boundaries and high throughput-per-dollar ratio. At its default reasoning capacity, it successfully isolates personal privacy variables without over-tokenizing non-human entities like pets or inanimate objects.
- gpt-5.4-mini **(The Guard Auditor):** Chosen for Stage 2 because of its strict compliance with structured JSON schemas and tool-calling validation parameters. Its distinct architectural lineage makes it an optimal independent validator to cross-examine Gemini's output without duplicating systemic bias configurations.

3. PROMPT DESIGN FRAMEWORK & JUSTIFICATION

To extract implicit privacy hazards reliably from highly unnormalized consumer product reviews, the linguistic interface of both our text-rewriter (Stage 1 Annotator) and our guard-verifier (Stage 2 Auditor) must be structured around proven prompt engineering methodologies. Rather than relying on trial-and-error text manipulation, our prompt architecture is engineered around three pillars: System-User Role Decoupling, Behavioral Constraint Scaffolding, and Schema-Enforced Structural Grounding.

3.1 System-User Message Decoupling and Persistent Scaffolding

Our pipeline establishes a rigid, hierarchical separation between application-level behaviors and session-specific text variables by leveraging the native API separation of system messages and user messages.

According to structural prompt optimization literature, splitting prompts into a bilevel optimization hierarchy prevents the model from suffering from task-agnostic contextual drift (Gozzi & Di Maio, 2024). Modern Large Language Models encode this instruction hierarchy via dedicated Chat Markup Languages that explicitly separate text inputs into isolated role environments. Defining foundational behavioral rules inside an invariant system message acts as a persistent constraint layer that dictates how the model processes information, balancing data accuracy with strict behavioral stability across structurally volatile target texts (Gozzi & Di Maio, 2024; Wang et al., 2025).

- **Application to Stage 1 (Annotator):** The complete definition of our custom tagset (<MINOR_AGE>, <FAM_KIN>, etc.) and our linguistic rules are hardcoded inside the persistent system instruction. The changing product review text (t) is passed exclusively as an isolated user token string. This configuration ensures that gemini-3.5-flash gives system-level instructions higher priority, preventing complex user syntax from accidentally hijacking the model's extraction logic.
- **Application to Stage 2 (Auditor):** The verification criteria and error types (TEXT_MUTATION, OVER_ANNOTATION) are housed entirely within the system prompt. The raw and annotated text blocks are submitted inside distinct user fields separated by punctuation boundaries (RAW_TEXT:\n{raw_text}). This structure allows the model to analyze data streams side-by-side without confusing internal review text with system commands.

3.2 Persona Assignment and Constraint Scaffolding

Both prompts explicitly employ **Behavioral Persona Framing** (e.g., *"You are a text processing engine"* and *"You are a quality assurance auditor"*). Research into role-playing prompt frameworks confirms that assigning targeted personas to language models activates specialized latent semantic tracks, which significantly yields lower defect rates when handling niche domain requirements (Gozzi & Di Maio, 2024).

Furthermore, we structurally insulate the pipeline against **Cognitive Inertia**—the tendency of language models to rigidly default to standard pre-training assumptions even when explicitly instructed to do otherwise (Wang et al., 2025). To mitigate cognitive inertia, our prompts are designed around **Negative Constraint Scaffolding** (Gozzi & Di Maio, 2024; Wang et al., 2025). We do not simply tell the models what to find; we define explicit exclusions using uppercase delimiters (*"Never tag ages or schools belonging to animals"*; *"ABSOLUTE WORD PRESERVATION: Do not fix typos"*). Explicitly restricting boundaries prevents the model from defaulting to broad, standard NER habits, ensuring strict alignment with our custom guidelines.

3.3 Schema Enforcers and Logical Reasoning

For the Stage 2 Auditing agent, our prompt architecture mandates a strict object-oriented payload by passing a Pydantic class constraint directly into the API configuration block.

- **Mitigating Evaluator Bias:** As highlighted in modern algorithmic evaluation studies, unconstrained multi-agent judging loops are highly susceptible to cognitive distortions,

such as the *halo effect* (over-estimating accuracy based on surface fluency) and *anchoring bias* (clinging to initial generation patterns) (Gozzi & Di Maio, 2024; Liang et al., 2023).
- **Forcing Analytical Friction via Explanations:** By forcing gpt-5.4-mini to populate a structured JSON schema that requires an explicit error_type and a mandatory auditor_reason, we eliminate open-ended conversational drift (Gozzi & Di Maio, 2024). Enforcing a structured output format restricts the model's generative token space, forcing the underlying decoder to follow rigorous logical reasoning steps before issuing a final validation status (Gozzi & Di Maio, 2024; Wang et al., 2025). This ensures that every flagged error corresponds directly to an actionable, categorized diagnostic trace.

4. COMPARATIVE ANALYSIS OF CONTRIBUTIONS

To contextualize the operational and theoretical advancements of our proposed multi-agent framework, we systematically compare our contributions against the state-of-the-art literature in Named Entity Recognition (NER), Relation Extraction (RE), and text anonymization previously analyzed (Efremova et al., 2015; Gillette et al., 2022; Mancera et al., 2025; Oliveira et al., 2021). Our approach introduces significant improvements in three core dimensions: architectural scaling, semantic granularity, and domain-specific noise mitigation.

4.1 Architectural Scaling and Human Optimization

Traditional pipelines for extracting privacy attributes or kinship graphs rely on manually constructed rule frameworks or resource-intensive small-model training configurations (Efremova et al., 2015; Gillette et al., 2022; Oliveira et al., 2021). As a result, scaling these methods to large-scale, unstructured datasets introduces severe processing bottlenecks.

- **Departure from Rule-Based Brittleness:** Oliveira et al. (2021) implemented Local Grammars and Cascades of Transducers to extract binary kinship predicates from text (Oliveira et al., 2021). While linguistically rigorous, their lines of deterministic grammar graphs cannot adapt to the dense stylistic variations, slang, and structural fragmentation typical of modern e-commerce product reviews. By leveraging advanced foundation models, our framework bypasses manual grammar generation entirely, utilizing zero-shot contextual reasoning to adapt to conversational text fluently.
- **Mitigation of Annotator Bias and Data Imbalance:** In their specialized minor-detection corpus, Gillette et al. (2022) encountered a severe 1:4 minor-to-adult class imbalance across 2,534 textual sequences (Gillette et al., 2022). To compensate, they were forced to implement synthetic data injection (via template-based name substitutions) to train a fine-tuned BERT architecture (Gillette et al., 2022). Furthermore, their annotators defaulted to an "adult" tag when clear age indicators were absent, causing the

downstream model to overly generalize the majority class and achieve a low 51% precision for adult entities (Gillette et al., 2022). Our dual-agent auditing architecture directly targets this vulnerability; by positioning `gpt-5.4-mini` as an adversarial critic to cross-examine `gemini-3.5-flash`, the pipeline acts as an automated validation barrier that flags and corrects systematic model biases before they can pollute the dataset (Liang et al., 2023).

- **Cognitive Optimization of Human-in-the-Loop (HITL) Curation:** While Efremova et al. (2015) utilized an expert web interface to establish a baseline consensus on 1,005 historical notary relationships, their approach required unguided manual extraction of candidate name pairs and relationship descriptors (Efremova et al., 2015). Scaling this unguided review format to a 20,000+ row dataset introduces massive labor overhead and cognitive fatigue. Our pipeline optimizes human interaction by forcing the auditing agent to output a structured JSON schema detailing the exact `error_type` and a concise, natural-language explanation for every flagged contradiction. This diagnostic layer narrows human focus from a generalized "search-and-debug" task to a targeted correction loop, allowing our framework to achieve gold-standard data validation across 20,000 rows at a fraction of the manual cost.

4.2 Semantic Granularity and Privacy Risk Profiling

Text anonymization methodologies are frequently constrained by flat, standardized entity schemas that overlook the complex, implicit data leaks found in personal narratives (Mancera et al., 2025).

- **Beyond Flat PII Masking:** The PBa-LLM framework presented by Mancera et al. (2025) demonstrated that text anonymization via Large Language Models can successfully safeguard sensitive information without damaging the mathematical utility of downstream transformer embeddings (Mancera et al., 2025). However, their privacy-preserving layer was strictly confined to flat Personally Identifiable Information (PII) categories, specifically extracting only Names (`PER`) and Geographic Locations (`LOC`) across resume profiles (Mancera et al., 2025). Our framework expands upon this objective by mapping a multi-dimensional, context-dependent privacy risk matrix. By tracking minor-specific data lineages (`MINOR_AGE`, `MINOR_EDU`), sensitive gender-revealing physiology (`GEN_PHYS`), and deep household relational networks (`FAM_KIN`), we demonstrate that generative models can isolate sophisticated, implicit identity tracks that standard, off-the-shelf PII scrubbers completely miss.

4.3 Domain-Specific Noise Mitigation

Unstructured text collections contain distinct variations of linguistic noise that require explicit semantic safety nets.

- **The E-Commerce Noise Challenge:** The historical legal registries and narrative corpora analyzed in prior literature naturally filter out non-human noise variables

(Efremova et al., 2015; Gillette et al., 2022). Efremova et al. (2015) operated on official notary documents where mentions of family boundaries were guaranteed to belong to real human citizens (Efremova et al., 2015). Conversely, consumer e-commerce data is inherently noisy; product reviews frequently describe the ages and development milestones of domestic animals or detail the character lore of fictional merchandise.

- **The Human Child Constraint:** To resolve this domain vulnerability, we introduce **The Human Child Constraint** as a core prompt instruction and an auditing filter. While rule-based frameworks like Oliveira et al. (2021) struggle with semantic ambiguity, our cascading prompt architecture ensures the extraction models distinguish real-world human minor vulnerabilities from background consumer noise (Oliveira et al., 2021). This insulates the dataset from over-tokenization and prevents non-risk variables from degrading the final data quality.

4.4 Comprehensive Cross-Literature Synthesis

Dimension	**Our Proposed Framework**	**Gillette et al. (2022)**	**Mancera et al. (2025)**	**Oliveira et al. (2021)**	**Efremova et al. (2015)**
Core Technical Engine	Generative Foundation Models (gemini-3.5-flash / gpt-5.4-mini)	Fine-Tuned Small Transformers (BERT-base-uncased)	General-Purpose LLMs vs. Microsoft Presidio	Rule-Based Local Grammars (Unitex Toolkit)	Sequential Probabilistic Models (Linear SVM & HMM)
Dataset Scale	**20,000+ Sequences (Amazon E-Commerce Reviews)**	2,534 Sequences (GlobalGiving Corpus)	24,000 Candidate Resume Profiles (FairCVdb)	Matthew Chapter 1 (Biblical Lineage Dataset)	347 Labeled Historical Notary Acts

Target Privacy Tagset	MINOR_AGE, MINOR_EDU, GEN_NOUN, GEN_PHYS, FAM_KIN	Binary Classification: MINR (Minor), ADLT (Adult)	Flat Standard PII: PER (Person), LOC (Location)	Relational Parental Predicates: Fatherhood/Motherhood	Kinship Triples: Marriage, Parent-Child, Sibling, Widow
Verification & Adjudication	**Multi-Agent Cross-Family Audit + Row-Level HITL Fix**	Ad-hoc Human Verification Sample Review	Automated Text Masking Transform Script (N)	Manual Local Grammar Rule Cascade Modification	Expert Web-Interface Double Annotation Consensus
Extraction Format	Absolute character offsets via inline text rewriting and regex parsing	Token-Level Word Chunk Sequence Tagging (BIO Format)	Latent State Vector Space Character Masking (t^{-})	Inline XML Concatenation Markup String Tags	Pairwise Word-Tag Index Tracking (BIO Mapping)
Domain Noise Isolation	**Yes** (Strict programmatic separation of pets, objects, and fiction)	No (Assumed structurally absent from raw narrative corpus)	No (Standardized resume layout constraints)	Partially (Filters figurative linguistic parentship)	No (Standardized legal demographic registry templates)

Dimension	**NuNER / Bogdanov et al. (2024)**

Core Technical Engine	LLM-annotated NER pretraining pipeline using GPT-3.5-generated entity annotations to further pre-train a compact RoBERTa-base encoder model called NuNER.
Dataset Scale	1.35M C4 sentences annotated by GPT-3.5, filtered to 1M annotated sentences with 4.38M entity annotations across roughly 200k unique concepts.
Target Privacy Tagset	Broad open-domain entity, concept, and idea labels; not privacy-specific and not limited to standard PII categories.
Verification & Adjudication	Automatic filtering and downstream NER benchmark evaluation; no dedicated cross-model audit or row-level human adjudication pipeline.
Extraction Format	LLM-generated entity/concept strings with span positions recovered through exact string matching rather than direct LLM-produced character offsets.
Domain Noise Isolation	Partially; broad web-domain entity diversity is encouraged, but the system does not define domain-specific filters such as separating human minors from pets, objects, or fictional entities.

5. DETAILED PROMPT STRUCTURE AND ARCHITECTURE

To transition our prompt engineering framework from high-level behavioral constraints to a production-ready deployment, we formalize the exact structural content of our system and user prompts. The configuration of these instruction vectors is designed around an optimized text-to-text transformation syntax for Stage 1 (Annotation) and an object-oriented structural schema constraint for Stage 2 (Auditing).

5.1 Stage 1: Primary Extraction & Annotation Engine Prompts

A. Core Prompt Payload

System Prompt

Role & Core Persona:

You are a deterministic, context-aware privacy metadata extraction engine. Your sole task is to rewrite the user's input sequence by injecting explicit, inline structural tags around text segments that disclose implicit or explicit household privacy metrics. You return the input verbatim with the sole exception of the injected tags.

Target Tagset Definitions:

1. <MINOR_AGE>: Numerical ages, developmental milestones, or age brackets of a REAL, LIVING

human child under 18 (e.g., "3yo", "toddler", "newborn", "as a minor myself"). Also see the Compound and Priority rules below — a count/kinship noun that denotes the reviewer's own child under 18 (e.g., "twins") is MINOR_AGE, not FAM_KIN.
2. <MINOR_EDU>: Academic classifications EXCLUSIVE to human minors (e.g., "kindergarten", "5th grade", "middle schooler", "high school freshman"). Never tag "college", "university", "trade school", or any pet/animal training program.
3. <GEN_NOUN>: Gendered identity nouns that anchor the REVIEWER or their DIRECT romantic partner (e.g., "husband", "wife", "hubby", "fiancée", "as a mom of two"). Never tag third-party gendered nouns unrelated to the reviewer's household (e.g., "the female cashier").
4. <GEN_PHYS>: Biological/medical states BOUND TO the reviewer's or partner's sex (e.g., "breastfeeding", "postpartum", "third trimester", "miscarriage"). Never tag conditions that are not gender-specific (e.g., "chest pain", "hair loss").
5. <FAM_KIN>: Kinship markers delineating the reviewer's active household and family matrix (e.g., "mother-in-law", "twin sister", "nephew", "granddaughter"). Adult children are tagged FAM_KIN (e.g., "my adult son") but receive NO minor tags. See the Priority rule for kinship nouns that denote a minor.

Span, Compound & Priority Rules:
- SPAN STRATEGY: Tag the ENTIRE noun phrase that carries the implicit meaning, including modifiers that anchor the risk, but exclude trailing punctuation.
  Right: "<MINOR_EDU>5th grade</MINOR_EDU>". Wrong: "in <MINOR_EDU>5th</MINOR_EDU> grade".
- DEMOGRAPHIC COMPOUNDS: When an age modifier is directly attached to a gendered noun representing a minor, DO NOT split the span. Tag the whole phrase as a single MINOR_AGE.
  Correct: "<MINOR_AGE>16-year-old girl</MINOR_AGE>".
  Incorrect: "<MINOR_AGE>16-year-old</MINOR_AGE> <GEN_NOUN>girl</GEN_NOUN>".
- FLAT NER PRIORITY: When one word or compact phrase is BOTH a kinship relation AND a minor attribute and cannot be cleanly separated, prioritize the minor attribute (MINOR_AGE) over FAM_KIN.
  Cleanly separable: "my <MINOR_AGE>3yo</MINOR_AGE> <FAM_KIN>son</FAM_KIN>".
  Not separable, minor context: "my <MINOR_AGE>stepson</MINOR_AGE> is in middle school" → "stepson" is MINOR_AGE (not FAM_KIN) because context fixes him under 18.
  Not separable: "my <MINOR_AGE>twins</MINOR_AGE>" (when under 18).

Inviolable Constraints:
- THE CHARACTER PRESERVATION MANDATE: Do not modify, fix, add, or delete any original characters, typos, punctuation, or spelling. Output must match the input character-for-character, with the SOLE exception of injected opening/closing tags. Do NOT expand "12yo" into "12-year-old".
- THE HUMAN CHILD CONSTRAINT: Disregard non-human entities. Never tag ages, milestones, or schooling of pets, animals, inanimate objects, brands, or vintage items (e.g., "my 2yo puppy", "dog training school", "my 5-year-old car").
- THE REVIEWER ANCHOR RULE: Only tag an entity that establishes an ACTIVE, real-world privacy footprint in the reviewer's current household. Never tag hypothetical consumer profiles, recommendation targets, gift-recipient suggestions, or abstract examples (e.g., in "Perfect gift for a wife", "wife" must NOT be tagged).

- THE HISTORICAL SELF-REFERENCE EXCLUSION: Never tag the reviewer's OWN childhood recalled in the past tense, as it describes an adult's history and poses no live minor-privacy risk (e.g., "When I was a teenager 20 years ago…" → tag nothing). Note this differs from a present-tense "as a minor myself", which IS MINOR_AGE.
- THE PREGNANCY-LOSS RULE: A miscarriage or pregnancy loss is tagged GEN_PHYS but NEVER MINOR_AGE — do not emit a phantom minor tag for a child that was not born.

User Prompt
[INPUT_SEQUENCE_INGESTION]
Review Text: """{raw_review_text}"""

B. Scholarly Justification

The layout of the Stage 1 prompt avoids conversational prompt drift by applying structured operational scaffolding. The division between the behavioral rule matrix (System Prompt) and the raw string block (User Prompt) utilizes native Chat Markup Language API boundaries to ensure invariant execution. This hierarchical insulation guarantees that the instruction set maintains higher execution priority over messy, unstructured user texts containing conversational noise or potential prompt injection attempts (Gozzi & Di Maio, 2024; Wang et al., 2025).

Furthermore, the explicit use of uppercase structural declarations (*"THE CHARACTER PRESERVATION MANDATE"*) serves as negative constraint scaffolding. This mechanism forces the foundation model's decoder away from standard pre-training assumptions—such as automatically correcting typos or punctuation layout—thereby minimizing text mutation failures caused by model cognitive inertia (Wang et al., 2025).

5.2 Stage 2: Independent Cross-Family Guard Auditor Prompts

A. Core Prompt Payload

System Prompt
Role & Core Persona:
You are a quality assurance data validation auditor. You are given two parameters: a pristine, raw consumer review (RAW_TEXT) and an inline-tagged version of that same review (ANNOTATED_TEXT).
Execute a side-by-side structural and semantic comparison and issue a FAIL if ANY rule below is triggered. Select the single DOMINANT error_type.

FAIL conditions:
1. TEXT_MUTATION: ANNOTATED_TEXT adds, deletes, or alters any original character, typo, punctuation, or spelling outside the XML tags (e.g., "12yo" expanded to "12-year-old").
2. INCORRECT_TAG: A tag is applied to a non-human entity, violating the Human Child Constraint (e.g., "<MINOR_AGE>1yo</MINOR_AGE> cat", "<MINOR_EDU>puppy school</MINOR_EDU>").
3. OVER_ANNOTATION: A tag is applied where no active reviewer-anchored privacy footprint exists.
This covers (a) hypothetical/gift/recommendation personas (Reviewer Anchor Rule), (b) the

reviewer's OWN past childhood in the past tense (Historical Self-Reference), and (c) non-gender-specific medical conditions wrongly tagged GEN_PHYS (e.g., "chest pain", "hair loss").
4. MISSING_TAG: A reviewer-anchored household relation, minor age/milestone, minor education tier, gender noun, or gender-specific physiology is present in RAW_TEXT but left untagged.
5. WRONG_LABEL: The span is a real, taggable entity but carries the wrong category (e.g., "wife" tagged FAM_KIN instead of GEN_NOUN; an adult child tagged MINOR_AGE instead of FAM_KIN; a miscarriage tagged MINOR_AGE instead of GEN_PHYS).
6. WRONG_SPAN: The label is correct but the boundary is wrong. This includes (a) trailing punctuation or a missing anchoring modifier ("in <MINOR_EDU>5th</MINOR_EDU> grade" instead of "<MINOR_EDU>5th grade</MINOR_EDU>"); (b) a Demographic Compound that was split ("<MINOR_AGE>16-year-old</MINOR_AGE> <GEN_NOUN>girl</GEN_NOUN>" instead of one "<MINOR_AGE>16-year-old girl</MINOR_AGE>"); (c) a Flat-NER-priority violation where a kinship noun denoting a minor was tagged FAM_KIN instead of MINOR_AGE ("stepson"/"twins" under 18).

Output Requirement:
Return your evaluation strictly through the requested JSON schema object. If status is FAIL, select the single dominant error_type and write a concise, 1-sentence auditor_reason naming the exact span and the rule violated. If status is PASS, set error_type to "NONE" and leave auditor_reason empty.

User Prompt
[AUDIT_VALIDATION_BLOCK]
RAW_TEXT: """{raw_review_text}"""
ANNOTATED_TEXT: """{tagged_annotator_output}"""

B. Scholarly Justification

The execution of the Stage 2 guard prompt is anchored in cross-family model collaboration to break the internal validation echo chambers common in standalone LLM deployments (Liang et al., 2023). By configuring gpt-5.4-mini to operate under strict JSON schema constraints via native API object-parsing hooks, we introduce an essential layer of analytical friction. Forcing the model to map its evaluation to an explicit, restricted key structure (status, error_type, auditor_reason) severely limits open-ended conversational drift and uncritical surface fluency approvals (Gozzi & Di Maio, 2024). Requiring a structured, categorized error reason forces the underlying decoder to execute a rigorous multi-step reasoning trace over the string sequences before finalizing its classification state, ensuring that the human-in-the-loop queue receives highly deterministic diagnostic data (Gozzi & Di Maio, 2024; Wang et al., 2025).

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